PHP LAB RECORD

Programs 1–10 | Questions, Source Code and Sample Outputs

Clean, white-themed reference document

Note: Outputs are illustrative. Programs involving forms, cookies, sessions, file uploads or MySQL should be executed through a local web server such as XAMPP. Save each PHP program with the filename indicated in its notes.

Program 1

Question: Write a PHP program to generate the Fibonacci series.

Program

```
<?php
$a = 0;
$b = 1;
echo "Fibonacci Series:<br>";
echo $a . "<br>" . $b . "<br>";
for ($i = 1; $i <= 8; $i++) {
    $c = $a + $b;
    echo $c . "<br>";
    $a = $b;
    $b = $c;
}
?>
```

Sample Output

```
Fibonacci Series:
0
1
1
2
3
5
8
13
21
34
```

Program 2

Question: Write a PHP program to swap two numbers using (a) pass by value and (b) pass by reference.

Program

```
<?php
// Pass by value
function swapValue($a, $b) {
    $temp = $a; $a = $b; $b = $temp;
    echo "Inside function: a=$a, b=$b<br>";
}
$x = 10; $y = 20;
echo "Before pass by value: x=$x, y=$y<br>";
swapValue($x, $y);
echo "After pass by value: x=$x, y=$y<br><br>";

// Pass by reference
function swapReference(&$a, &$b) {
    $temp = $a; $a = $b; $b = $temp;
}
$x = 10; $y = 20;
echo "Before pass by reference: x=$x, y=$y<br>";
swapReference($x, $y);
echo "After pass by reference: x=$x, y=$y";
?>
```

Sample Output

```
Before pass by value: x=10, y=20
Inside function: a=20, b=10
After pass by value: x=10, y=20

Before pass by reference: x=10, y=20
After pass by reference: x=20, y=10
```

Program 3

Question: Write a PHP program to demonstrate array-related functions.

Note: Note: print_r() displays array indexes and formatting in the browser.

Program

```
<?php
$fruits = array("Apple", "Banana", "Cherry");
echo "Original array: "; print_r($fruits);
array_push($fruits, "Date", "Elderberry");
echo "<br>After array_push: "; print_r($fruits);
$removed = array_pop($fruits);
echo "<br>After array_pop (removed $removed): "; print_r($fruits);
$removed = array_shift($fruits);
echo "<br>After array_shift (removed $removed): "; print_r($fruits);
array_unshift($fruits, "Apricot");
echo "<br>After array_unshift: "; print_r($fruits);
$combined = array_merge($fruits, array("Carrot", "Beetroot"));
echo "<br>After array_merge: "; print_r($combined);
echo "<br>After array_reverse: "; print_r(array_reverse($combined));
echo "<br>Total elements: " . count($combined);
?>
```

Sample Output

```
Original array: Apple, Banana, Cherry
After array_push: Apple, Banana, Cherry, Date, Elderberry
After array_pop (removed Elderberry): Apple, Banana, Cherry, Date
After array_shift (removed Apple): Banana, Cherry, Date
After array_unshift: Apricot, Banana, Cherry, Date
After array_merge: Apricot, Banana, Cherry, Date, Carrot, Beetroot
After array_reverse: Beetroot, Carrot, Date, Cherry, Banana, Apricot
Total elements: 6
```

Program 4

Question: Write a PHP program to count vowels and consonants in a given string.

Program

```
<?php
$str = "The future belongs to those who prepare for its today!";
$vowels = 0; $consonants = 0; $spaces = 0;
$str = strtolower($str);
for ($i = 0; $i < strlen($str); $i++) {
    $ch = $str[$i];
    if (strpos("aeiou", $ch) !== false) {
        $vowels++;
    } elseif ($ch == " ") {
        $spaces++;
    } elseif (ctype_alpha($ch)) {
        $consonants++;
    }
}
echo "Input: $str<br>";
echo "Vowels: $vowels<br>";
echo "Consonants: $consonants<br>";
echo "Spaces: $spaces";
?>
```

Sample Output

```
Input: the future belongs to those who prepare for its today!
Vowels: 17
Consonants: 31
Spaces: 8
```

Program 5

Question: Write a PHP program to read employee details and display them using a form.

Note: Save as employee.php and run through localhost. The output shown is after sample form submission.

Program

```
<!DOCTYPE html>
<html><body>
<h2>Enter Employee Details</h2>
<form method="post" action="">
Name: <input type="text" name="emp_name" required><br><br>
Employee ID: <input type="text" name="emp_id" required><br><br>
Department: <input type="text" name="emp_department" required><br><br>
```

```
Email: <input type="email" name="email" required><br><br>
<input type="submit" name="submit" value="Submit">
</form>
<?php
if (isset($_POST['submit'])) {
    echo "<h2>Employee Details Entered:</h2>";
    echo "Name: " . htmlspecialchars($_POST['emp_name']) . "<br>";
    echo "Employee ID: " . htmlspecialchars($_POST['emp_id']) . "<br>";
    echo "Department: " . htmlspecialchars($_POST['emp_department']) . "<br>";
    echo "Email: " . htmlspecialchars($_POST['email']) . "<br>";
}
?>
</body></html>
```

Sample Output

```
Employee Details Entered:
Name: Sai Mukesh
Employee ID: EMP101
Department: Computer Science
Email: sai@example.com
```

Program 6

Question: Create a student registration form using text box, radio button, checkbox, select and submit button, and display the user input values in a new PHP page.

Note: Create student.php and display.php in the same folder and run through localhost.

Program

```
<!-- Save as student.php -->
<!DOCTYPE html><html><body>
<h2>Student Registration</h2>
<form method="post" action="display.php">
Name: <input type="text" name="name" required><br>
DOB: <input type="date" name="dob"><br>
Email: <input type="email" name="email"><br>
Gender: <input type="radio" name="gender" value="Male"> Male
<input type="radio" name="gender" value="Female"> Female<br>
Courses:<br>
<input type="checkbox" name="course[]" value="Maths"> Maths<br>
<input type="checkbox" name="course[]" value="Science"> Science<br>
<input type="checkbox" name="course[]" value="English"> English<br>
Department: <select name="department">
<option>Computer Science</option><option>Electronics</option><option>Mechanical</option>
</select><br>
<input type="submit" value="Register">
</form></body></html>

<!-- Save as display.php -->
<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    echo "Student Details Submitted<br>";
    echo "Name: " . htmlspecialchars($_POST["name"] ?? "") . "<br>";
    echo "DOB: " . htmlspecialchars($_POST["dob"] ?? "") . "<br>";
    echo "Email: " . htmlspecialchars($_POST["email"] ?? "") . "<br>";
    echo "Gender: " . htmlspecialchars($_POST["gender"] ?? "") . "<br>";
    echo "Department: " . htmlspecialchars($_POST["department"] ?? "") . "<br>";
    echo "Selected courses:<br>";
    foreach ($_POST["course"] ?? array() as $course) {
        echo htmlspecialchars($course) . "<br>";
    }
} else {
    echo "Form not submitted";
}
?>
```

Sample Output

```
Student Details Submitted
Name: Sai Mukesh
DOB: 2005-06-15
Email: sai@example.com
Gender: Male
Department: Computer Science
Selected courses:
Maths
Science
```

Program 7

Question: Write a PHP script to upload a file.

Note: Save index.html and upload.php in one folder. This introductory example does not include production-grade file type/size validation.

Program

```
<!-- Save as index.html -->
<!DOCTYPE html><html><body>
<form enctype="multipart/form-data" action="upload.php" method="post">
Select file: <input type="file" name="uploaded_file" required>
<input type="submit" value="Upload">
</form></body></html>

<!-- Save as upload.php -->
<?php
if (isset($_FILES["uploaded_file"]) &&
    $_FILES["uploaded_file"]["error"] === UPLOAD_ERR_OK) {
    $uploadDir = __DIR__ . "/uploads/";
    if (!is_dir($uploadDir)) mkdir($uploadDir, 0755, true);
    $fileName = basename($_FILES["uploaded_file"]["name"]);
    $destination = $uploadDir . $fileName;
    if (move_uploaded_file($_FILES["uploaded_file"]["tmp_name"], $destination)) {
        echo "The file " . htmlspecialchars($fileName) . " has been uploaded.";
    } else {
        echo "There was an error uploading the file.";
    }
} else {
    echo "Please select a file to upload.";
}
?>
```

Sample Output

```
The file photo.jpg has been uploaded.
```

Program 8

Question: Write a PHP script to demonstrate passing variables with cookies.

Note: Create setcookie.php and get-cookie.php in the same folder. Cookie values become available on a subsequent browser request.

Program

```
<!-- Save as setcookie.php -->
<?php
// setcookie() must be called before output.
setcookie("username", "Sai", time() + 3600, "/");
?>
<!DOCTYPE html><html><body>
<a href="get-cookie.php">Click here</a>
</body></html>

<!-- Save as get-cookie.php -->
<?php
if (isset($_COOKIE["username"])) {
    $user = $_COOKIE["username"];
    echo "Welcome back, " . htmlspecialchars($user) . "!";
} else {
    echo "Cookie 'username' is not set.";
}
?>
```

Sample Output

```
setcookie.php page:
Click here

After clicking the link (subsequent request):
Welcome back, Sai!
```

Program 9

Question: Write a PHP program to keep track of how many times a visitor has loaded the page.

Note: Refresh the page within the same session. The counter is session-based.

Program

```
<?php
session_start();
if (isset($_SESSION["count"])) {
    echo "Your session count: ";
    echo $_SESSION["count"] . "<br>";
    $_SESSION["count"]++;
} else {
    $_SESSION["count"] = 1;
    echo "Session does not exist";
}
?>
```

Sample Output

```
First page load:
Session does not exist

Second page load:
Your session count: 1

Third page load:
Your session count: 2

Fourth page load:
Your session count: 3
```

Program 10

Question: Discuss the steps to create, view and delete the records from a MySQL database using PHP.

Note: Run the SQL setup first, then students.php through localhost. This sample inserts Ravi and deletes record ID 1. For repeat testing, reset the table or adjust the ID.

Program

```
-- Run these SQL commands first in MySQL:
CREATE DATABASE college;
USE college;
CREATE TABLE students (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    marks INT
);

-- Save the following PHP code as students.php
<?php
// Step 1: Connect to the database
$conn = mysqli_connect("localhost", "root", "", "college");
if (!$conn) die("Connection failed: " . mysqli_connect_error());
echo "Database connected successfully<br><br>";

// Step 2: Create/insert a record
$sql = "INSERT INTO students (name, marks) VALUES ('Ravi', 85)";
if (mysqli_query($conn, $sql)) {
    echo "Record created successfully<br><br>";
} else {
    echo "Error creating record: " . mysqli_error($conn) . "<br>";
}

// Step 3: View records
echo "<h3>Student Records</h3>";
$result = mysqli_query($conn, "SELECT * FROM students");
while ($row = mysqli_fetch_assoc($result)) {
    echo "ID: " . $row['id'] . "<br>";
    echo "Name: " . htmlspecialchars($row['name']) . "<br>";
    echo "Marks: " . $row['marks'] . "<br><br>";
}

// Step 4: Delete record with ID 1
$id = 1;
$sql = "DELETE FROM students WHERE id = $id";
if (mysqli_query($conn, $sql)) {
    echo "Record deleted successfully<br>";
} else {
    echo "Error deleting record: " . mysqli_error($conn);
}
// Step 5: Close the connection
mysqli_close($conn);
```

```
?>
```

Sample Output

```
Database connected successfully

Record created successfully

Student Records
ID: 1
Name: Ravi
Marks: 85

Record deleted successfully
```